

Rotman

INTRO TO R PROGRAMMING

R Tutorial (RSM358) – Session 2 (Optional Materials)

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Website: <https://rmdal.github.io/r-intro-2026-fall/>



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Linear Regression - Housing Price & Clean Air

Obs: 506

- Manipulate data
 - Load data
 - Create new columns
 - Filter columns and rows
- Build models
 - Multiple linear regressions
- Report and graph
 - Plot a few graphs
 - Report regression results

1. price	median housing price, \$
2. crime	crimes committed per capita
3. nox	nitrous oxide, parts per 100 mill.
4. rooms	avg number of rooms per house
5. dist	weighted dist. to 5 employ centers
6. radial	accessibiliy index to radial hghwys
7. proptax	property tax per \$1000
8. stratio	average student-teacher ratio
9. lowstat	% of people 'lower status'

Choice 1: Use Only Base R packages

- Manipulate data
 - Load data ([read.csv\(\)](#))
 - Create new columns ([base R data frame manipulation](#))
 - Filter columns and rows ([base R data frame manipulation](#))
- Build models
 - Multiple regression ([lm\(\)](#) from stats library in R base)
- Report and graph
 - Base R plot system, [plot\(\)](#)
 - Base R [summary\(\)](#) function
- A Note on Predictive Analysis
 - Train and test (or validation) split
 - Predict on test data and obtain evaluation measures of interest

Choice 2: Use 3-party Packages (Optional)

- Manipulate data (tidyverse eco-system)
 - Load data (read_csv() from the readr)
 - Create new columns (mutate() from dplyr)
 - Filter columns and rows (select() and filter() from dplyr)
- Build models
 - Multiple regression (lm() from stats library in R base)
- Report and graph
 - Graph using ggplot2 and some of its extensions
 - Build a publication-ready table (huxreg() from huxtable library)

Load a CSV file

- Choice1: [read.csv\(\)](#) from Base R's utils library (load into dataframe)

```
read.csv(file)
```

```
e.g. hprice <- read.csv("hprice.csv")
```

- Choice 2: [read_csv\(\)](#) from [tidyverse](#)'s [readr](#) library (load into tibble/dataframe)

```
read_csv(file)
```

```
e.g. hprice <- read_csv("hprice.csv")
```

Data Frame Manipulation – Base R vs dplyr

Data Process Operation	Base R	dplyr
Create a new column variable (from other column variables)	<code>df\$z <- df\$x + df\$y</code> , or <code>df["z"] <- df["x"] + df["y"]</code> , or <code>transform()</code>	<code>mutate(df, z = x + y)</code>
Filter rows based on conditions	<code>df[which(x), , drop = FALSE]</code> , or <code>subset()</code>	<code>filter(df, x)</code>
Select column variables	<code>df[c("x", "y")]</code> , or <code>subset()</code>	<code>select(df, x, y)</code>
...	...	...

Source: <https://dplyr.tidyverse.org/articles/base.html>

Data Manipulation: dplyr basics

- Filter observations (rows): filter()

```
filter(my_dataframe, condition1, ...)  
e.g., hprice_reg <- filter(hprice, price > 20000)
```

- Create new variables: mutate()

```
mutate(my_dataframe, new_var1 = expression1, ...)  
e.g., hprice_reg <- mutate(hprice_reg, lprice = log(price))
```

- Select variables (columns): select()

```
select(my_dataframe, var1, ...)  
e.g., hprice_reg <- select(hprice_reg, lprice, rooms)
```

Data Manipulation: Data Pipe (%>%)

```
hprice_reg <- filter(hprice, price > 20000)
hprice_reg <- mutate(hprice_reg, lprice = log(price))
hprice_reg <- select(hprice_reg, lprice, rooms)
```



```
hprice_reg <- hprice %>%
  filter(price > 20000) %>%
  mutate(lprice = log(price)) %>%
  select(lprice, rooms)
```

Linear Regression

- Multiple regressions: [lm\(\)](#) from stats library in base R

```
my_model <- lm(y ~ x1 + x2, data)
```

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon_i$$

```
my_model <- lm(y ~ x1 + x2 + I(x1 * x2), data)
```

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon_i$$

- Regression result summary: `summary()`

Ref. <https://faculty.chicagobooth.edu/richard.hahn/teaching/FormulaNotation.pdf>

Report

- Choice 1: Summary table (Base R; [summary for lm\(\)](#))

```
summary(my_model)
```

- Choice 2: publication-ready table: [huxreg\(\)](#) from [huxtable](#) library

```
huxtable(my_model1, my_model2, ...)
```

Read the Linear Regression Report

Call:

```
lm(formula = lprice ~ lnox + rooms + I(rooms^2) +  
stratio, data = hprice_reg)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.67205	-0.11678	0.01795	0.11597	0.59801

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.17845	0.47124	27.965	< 2e-16 ***
lnox	-0.58449	0.04594	-12.724	< 2e-16 ***
rooms	-0.69766	0.14221	-4.906	1.30e-06 ***

...

...

I(rooms^2)	0.07211	0.01129	6.385	4.29e-10 ***
stratio	-0.03929	0.00426	-9.223	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1833 on 448 degrees of freedom
Multiple R-squared: 0.6188, Adjusted R-squared: 0.6154
F-statistic: 181.8 on 4 and 448 DF, p-value: < 2.2e-16

Interpret Linear Regression Result ($\hat{\beta}_i$)

- $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ (x_1 is continuous)
- $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ (x_1 is categorical, say, 0 or 1)
- $\log(\hat{y}) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ (y is log-transformed)
- $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \log(x_1) + \hat{\beta}_2 x_2$ (x_1 is log-transformed)
- $\log(\hat{y}) = \hat{\beta}_0 + \hat{\beta}_1 \log(x_1) + \hat{\beta}_2 x_2$ (y and x_1 are log-transformed)
- $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2$ (an interactive term)

Ref. <https://stats.oarc.ucla.edu/other/mult-pkg/faq/general/faqhow-do-i-interpret-a-regression-model-when-some-variables-are-log-transformed/>

A Note on Predictive Analysis

- Causal vs predictive analysis
- Training and test (validation) data split
- Three Steps
 1. randomly split the data into training and test set.
 2. train/estimate a model on training set.
 3. Evaluate the estimated model on test set, i.e., predict on the test set, and obtain evaluation measures of interest.